



# Graphs of linear equations

## What to do

- What is a linear function?
- Explain how you would graph the line with the equation  $3x - 4y = 12$ .
- Use the gradient and  $y$ -intercept to sketch graphs of the following equations.
  - $y = 4x + 2$
  - $y = -2x + 5$
  - $y = \frac{2}{3}x - 1$
  - $y = 4 - 3x$
- Use  $x$ - and  $y$ -intercepts to sketch the following equations.
  - $y + x = 4$
  - $y = 5 - x$
  - $3y + 2x = 12$
  - $3y - 2x = 6$
  - $4x - 2y = 8$
  - $\frac{x}{2} + \frac{y}{3} = 1$
- Find the value of  $b$  given that the point  $(1, 5)$  lies on the line  $y = b - 3.5x$ .
  - What is the  $x$ -intercept?
- If a linear graph has an  $x$ -intercept of 4 and a  $y$ -intercept of  $-2$ , is the gradient positive or negative? Explain.
  - If a linear graph has an  $x$ -intercept of  $-2$  and a  $y$ -intercept of  $-3$ , is the gradient positive or negative? Explain.
- What is the slope of a horizontal line? Sketch a graph and verify your answer.
- What is the slope of a vertical line? Sketch a graph and verify your answer.
- Sketch graphs for each of the following linear equations.
  - $y = 2$
  - $-3 - y = 0$
  - $\frac{2}{3}x = -1$
- Graph each of the following functions within the given domains.
  - $f(x) = 2x + 1, -2 \leq x \leq 3$
  - $f(x) = 5 - x, -3 \leq x \leq 2$
  - $f(x) = -4x - 1, -2 < x < 2$
- Given the following graphs, state the range.
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